

Where Does Our Waste Go?

Tracking the Invisible Journey of Waste

Developer Brief — Page & Role Requirements

Purpose: A small functional academic web prototype to demonstrate tracking of a waste batch through Collection → Transportation → RTS/Transfer → Processing.

1. Login Page

- One common login page for all six roles: Admin, Collection, Transportation, RTS/Transfer, Processing, and Head Officer.
- After login, redirect the user to the dashboard/page appropriate to their role.

2. Admin

- **Admin Dashboard:** total batches, basic counts by current stage/status, and recent updates.
- **Create Waste Batch:** automatically generated Batch ID; waste type (Wet/Dry); optional quantity; source/location; date/time; initial status.
- **QR generation:** create a unique QR linked to the batch tracking page.
- **Batch Management:** view/search batches by Batch ID and open complete journey/history.
- **User Management:** manage the six user accounts/roles.
- **CSV Export:** export batch and stage tracking records.

3. Collection User

- **Dashboard:** show batches relevant to Collection.
- **Update Page:** add stage, date/time, manually entered location, short note/status, and optional quantity.
- **Permission:** can update only Collection-stage information.

4. Transportation User

- **Dashboard:** show batches relevant to Transportation.
- **Update Page:** add stage, date/time, manually entered location, short note/status, and optional quantity.
- **Permission:** can update only Transportation-stage information.

5. RTS / Transfer User

- **Dashboard:** show batches at the RTS/Transfer stage.
- **Update Page:** add stage, date/time, manually entered location, short note/status, and optional quantity.
- **Permission:** can update only RTS/Transfer-stage information.

6. Processing User

- **Dashboard:** show batches relevant to Processing.
- **Update Page:** add stage, date/time, manually entered location, short note/status, and optional quantity.
- After Processing is recorded, the journey can be shown as completed.

7. Head Officer

- **View-only Dashboard:** total batches, current stage/status, batch list, and complete journey.
- **Tracking:** search Batch ID and view all stage-wise updates.
- **Permission:** cannot add or edit stage updates.

8. Batch Tracking Page

- This is the main page opened by Batch ID search or QR scan.
- **Batch Details:** Batch ID, waste type, quantity if available, source, and current status.
- **Journey Timeline:** Collection → Transportation → RTS/Transfer → Processing.
- Each stage shows date/time, manually entered location, and status/note.
- Stages appear in chronological order.

9. QR Code

- Every newly created batch receives a unique QR code.
- Scanning the QR opens the corresponding Batch Tracking Page.
- QR should link to the Batch ID/tracking page rather than storing all waste data inside the QR.

10. Search

- **Batch ID → Search → Batch Tracking Page.**
- The same tracking page should open through QR scanning.

11. Overall User Flow

Step	Action
1	Admin creates Waste Batch
2	System generates unique Batch ID + QR
3	Collection user adds Collection update
4	Transportation user adds Transportation update
5	RTS user adds RTS/Transfer update
6	Processing user adds Processing update
7	Complete journey appears on Tracking Page
8	Head Officer views complete journey
9	Admin exports records as CSV